

Mathematics

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Chapter 1

Groups

1.1 Magmas

Definition 1.1. A magma is a pair $(S, *)$ with S a set and

$$* : S \times S \rightarrow S$$

an internal composition law.

Definition 1.2. A magma $(S, *)$ is said to be associative if $\forall a, b, c \in S, (a * b) * c = a * (b * c)$.

Definition 1.3. A magma $(S, *)$ is said to be commutative if $\forall a, b \in S, a * b = b * a$.

Definition 1.4. An identity element of a given magma $(S, *)$ is an element $e \in S$, such that $\forall a \in S, e * a = a * e = a$.

Theorem 1.5. *If a magma $(S, *)$ has an identity element, then it is unique.*

Proof. If $e, e' \in S$ are identity elements for $*$, then $e = e * e' = e'$. □

Notation 1.6. If $(S, *)$ is an associative magma with an identity element $e \in S$, then for all $a \in S$, we denote:

$$a^0 = e \quad \text{and} \quad a^n = \underbrace{a * \cdots * a}_{n \text{ times}} \quad (\text{for } n \geq 1).$$

Definition 1.7. Given a magma $(S, *)$ with an identity element $e \in S$. An element $a \in S$ is said to be invertible if there exists b such that $a * b = b * a = e$. b is then called an inverse of a .

Theorem 1.8. *If $(S, *)$ is an associative magma, then every invertible element $a \in S$ has a unique inverse, denoted by a^{-1} .*

Proof. Let $b, c \in S$ be two inverses of a . Then we have

$$b = b * (a * c) = (b * a) * c = c.$$

□

1.2 Groups

Definition 1.9. A group is an associative magma with an identity element, such that every element is invertible.